

703159 VU

Data-driven Analysis of Geodata

Session #02: Basics of Machine Learning

Dr. Dominik Amschwand, Prof. Jan Beutel

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CONTENT

1. Why ML? A motivating application
2. Definition and key components of ML:
Representation, Evaluation, Optimisation
3. Algorithm taxonomies
4. Bias–variance trade-off
5. Distance metrics
6. Curse of dimensionality
7. Five schools of ML
8. Applications of ML in the geosciences

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Key readings: Domingos [2012], [Chapters 1 and 5 of Goodfellow et al., 2016].

A whig history of data-driven permafrost distribution modelling

- ‘Whig history’: The past retold as inevitable prologue to the present, or ‘hindsight dressed up as destiny’.
- Classifying the terrain whether the subsurface is year-round frozen.
- Permafrost is invisible at the surface, subsurface information is scarce: Modelling is the tool of choice.
- Task: Link hidden subsurface state (frozen/non-frozen) to observable surface features.
- Two approaches: Process-oriented and empirical-statistical modelling

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

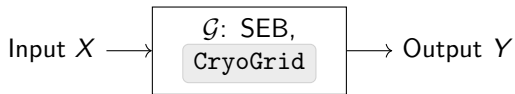
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Applications of ML in the geosciences

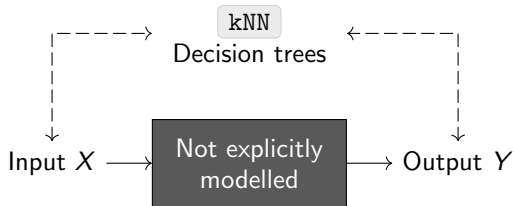
References

Process-oriented and statistical permafrost modelling



(a) Process-oriented modelling.

Processes explicitly modelled (albeit often simplified).



(b) Empirical-statistical modelling.

Process knowledge implicit via inductive bias.

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

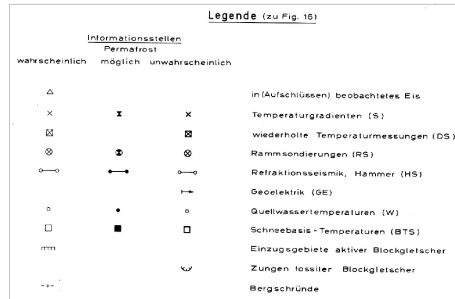
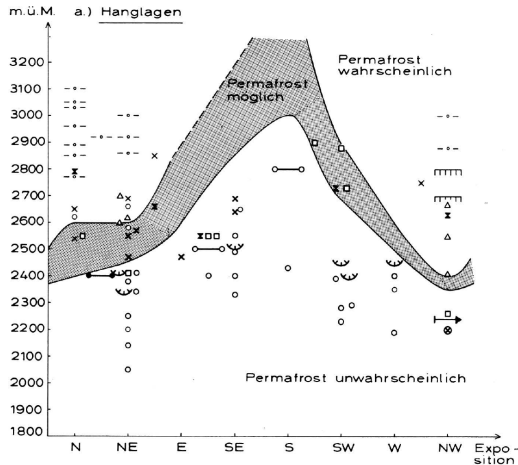
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Applications of ML
in the geosciences

References

Instance-based: Classify based on similarity and analogy

NN



Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias-variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Whig history. [Haeberli, 1975],

[Haeberli, 1975].

From instance- to rule-based learning

- Local evidence for absence/presence of permafrost (with uncertainty margin)
- **Instance-based learning:** *Keep all the data* and do nothing (“lazy learning”). Query for the most similar/closest point: nearest neighbour and k -nearest neighbours **kNN**
- **kNN**: non-parametric, flexible boundaries, simple implementation, but data-hungry, susceptible to CoD.
- “Closest” in terms of what? Geographic location, but also similar temperature, insolation, ... → Process understanding from domain knowledge is the key for generalisation.
- **Rule-based learning:** Derive rules from the data (generalize), then *forget data*.
- **Decision Tree** (with process-informed structure)

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

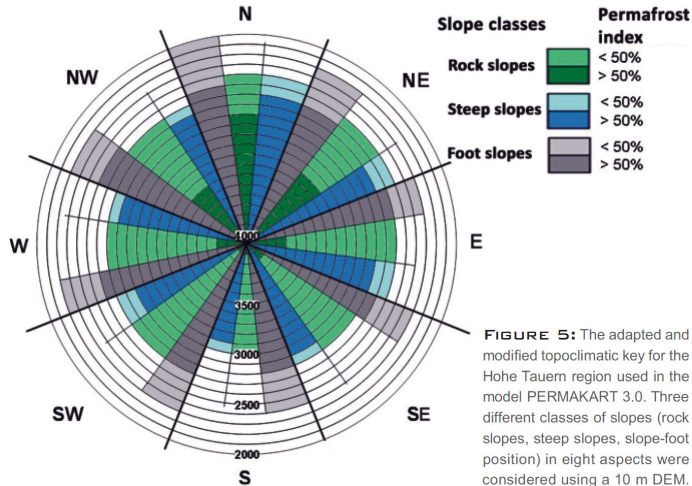
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Applications of ML
in the geosciences

References

[Domingos, 2018, p.179]; Borges [1944]: “To think is to forget differences, generalize, abstract.”

Rule-based permafrost prediction



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

[Ebohon & Schrott, 2008; Schrott et al., 2012; Otto et al., 2020].

Rule-based permafrost prediction using a decision tree

The 'rules of thumb' by Haeberli [1975] were subsequently refined, implemented in a GIS [Keller, 1992], and regionally adapted. Process knowledge is implicitly embedded.

Target: Permafrost?

$$\{0,1,2\} = \{\text{no PF, possible PF, likely PF}\}$$

[1. Slope class] (accounting for snow and terrain characteristics)

└ Rock slope

[2. Aspect] (accounting for insolation/radiation balance)

 $\vdash \{N, SE, E, \dots, NW\}$

[3. Elevation] (accounting for air temperature/turbulent fluxes)

— {<2400 m asl., 2400-2800 m asl., >2800 m asl.}:

--> {no PF, possible PF, likely PF}

— Steep slope

└ Foot slope

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Automation: Human $\rightarrow$ Computer

- The two approaches discussed rely heavily on hand-crafted features and rules (and predate the ML-in-geoscience era)
- Clear, interpretable structure, but labour-intensive to adapt to other mountain ranges

Can we automatise this? Yes (partly): Make it more strongly data-driven.

- Let the algorithm learn more from the data directly and infer the permafrost distribution.
- This is what machine learning does: Learns pattern from data with much less human interference, and discovers pattern (in huge data sets) that humans would not be able to.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Mark I Perceptron (1958)

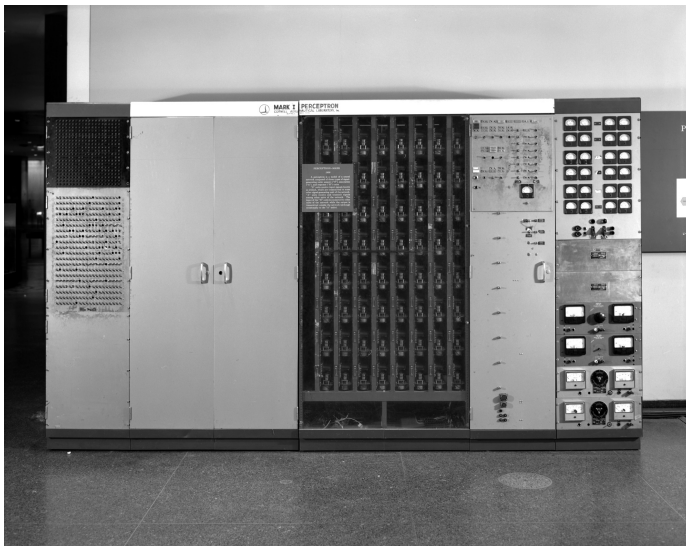


Image source.

NMAH-72-361 Mainframe Computer Components, Mark I Perceptron, Overall Front View.

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Key Components of ML

We assume a (causal or merely statistical) **relationship** between predictors X and a target Y :

$$Y = f(X) + \varepsilon, \quad (1)$$

where f is a fixed but unknown function, and ε is random noise independent of X (i.e. $\varepsilon \perp\!\!\!\perp X$).

Goal: estimate f from data. Two motivations:

- **Prediction:** Get $\hat{y}$ as close to y as possible to predict y for new, unseen x (we don't care what $\hat{f}$ looks like internally)
- **Inference:** Understand *how* X influences Y (interpretability matters)

The noise term ε sets a hard limit on prediction accuracy: the **irreducible error**.

Notation: estimated function/value $\hat{}$ (hat), 'true' unknown one without hat. [James et al., 2023].

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Key Components of ML

We want to find (“learn”) a function $\hat{f}$ with parameters θ and inductive bias b that produces an estimated output $\hat{y}$ that comes ‘close enough’ to the true y , i.e., such that

$$\hat{f}_b(X; \theta) = \hat{y} \approx y. \quad (2)$$

This means solving:

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(\underbrace{\hat{f}(X; \theta)}_{\hat{y}}, y), \quad (3)$$

where $\mathcal{L}(\hat{y}, y)$ is a **loss function** measuring how far $\hat{y}$ is from y .

Inductive bias b : the assumptions baked into the choice of $\hat{f}$ (e.g. “relationships are linear”). Every model has one: choosing $\hat{f}$ is choosing a bias.

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Key components of ML

We need to specify a form for $\hat{f}$ that a computer can handle (representation), an objective the computer optimises to determine which $\hat{f}$ is 'best' (evaluation), and a procedure by which the computer efficiently searches for $\hat{f}$ (optimisation):

- **Representation** — What machine-executable form can $\hat{f}$ take? (linear model, decision tree, neural network. . .)
- **Evaluation** — What is $\mathcal{L}$? (accuracy, RMSE, log-loss. . .)
- **Optimisation** — How does the computer find the best θ ? (gradient descent, grid search. . .)

All ML algorithms consist of 3 components:

Learning = Representation + Evaluation + Optimisation.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Representation: Data and algorithm must be machine-readable



Image generated by AI, based on *La Trahison des images* by René Magritte.

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Representation: What is machine-learnable?

Representation is the formal language in which the machine expresses its learning models and the data. Representation defines the **feature space** (what we feed in as X) and the **hypothesis space** $\mathcal{H}$ (what shape $\hat{f}$ can take).

Two inseparable choices:

- **Feature space:** What do we supply as X ? (raw measurements, derived indices, transformed variables. . .)
- **Hypothesis space:** What functional form can $\hat{f}$ take? (linear model, decision tree, neural network. . .)

[Domingos, 2012; James et al., 2023].

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Representation: What is machine-learnable?

The **hypothesis space** $\mathcal{H}$ is the set of all possible functions $\hat{f}$ (“hypotheses”) a learning algorithm can consider as candidates for the true f . A learner cannot find a solution outside $\mathcal{H}$. $\mathcal{H}$ is defined by the chosen model architecture (e.g., linear regression, neural network).

- **Linear regression:** $\mathcal{H}$ contains only linear functions (parametric, strong assumptions).
- **Nearest neighbour:** $\hat{f}$ is implicitly defined by the training data without a fixed functional form, but choice of distance metric crucial.

If the true f lies outside $\mathcal{H}$, no amount of data or optimisation can recover it, only the best approximation *within* $\mathcal{H}$.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Inductive Bias: Assumption-free modelling is impossible

A model trained on finite data must generalise to unseen cases. But infinitely many functions $\hat{f}$ fit any finite dataset perfectly. The algorithm must prefer some over others based on assumptions. This preference is its **inductive bias**.

Every model assumes something about the world:

- **Linear model:** Target Y responds proportionally to features X
- **Nearest neighbours:** Similar input features have similar output targets.

Inductive bias also enters before modelling: the **choice of features** X assumes those variables are the relevant ones.

The **No Free Lunch theorem** [Wolpert, 1996] formalises why bias is inescapable and why assumption-free models would not work in practice: No algorithm outperforms all others across all problems.

[Domingos, 2012; Wolpert, 1996].

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

EXAMPLE Permafrost prediction

Permafrost example: Choosing elevation, MAAT, and NDVI as features already encodes the belief that topography and climate — not, say, bedrock lithology — are the primary controls on permafrost. That is an inductive bias, and it (ideally¹) is a *scientific* choice, not a technical one.

Much of the work in applied ML is here — especially for domain scientists.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

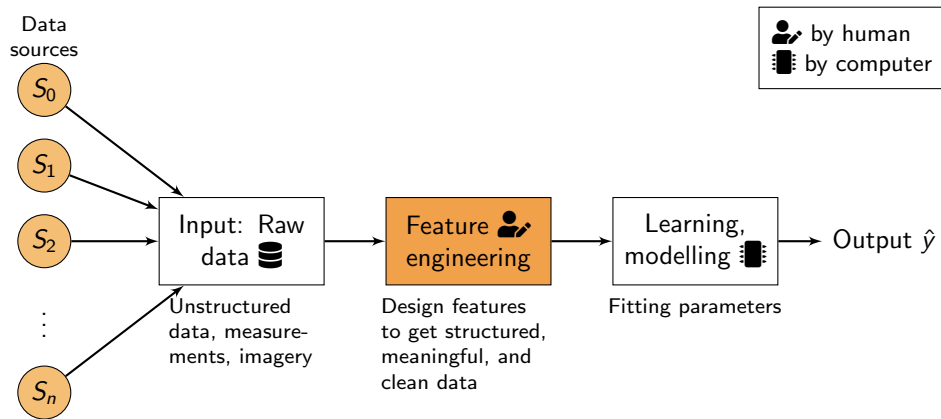
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Applications of ML in the geosciences

References

¹Technical aspects like data availability does play a role in practice.

Feature Engineering: Making data machine-readable



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

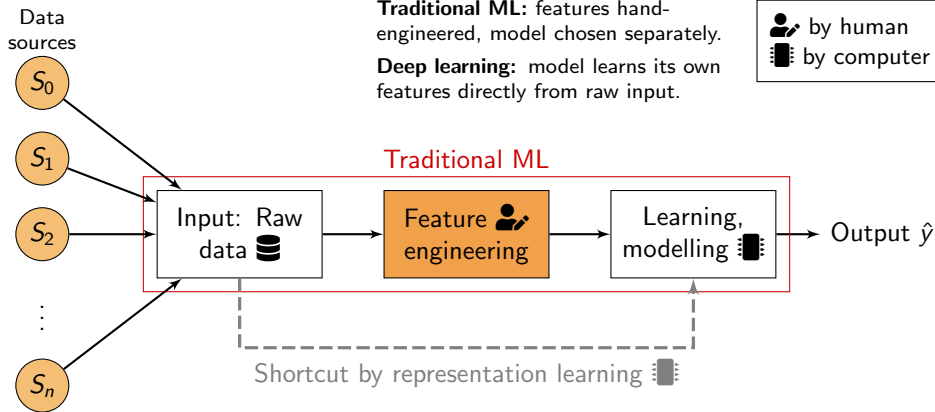
References

Representation learning skips feature engineering

Traditional ML: features hand-engineered, model chosen separately.
Deep learning: model learns its own features directly from raw input.

 by human
 by computer

Traditional ML



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

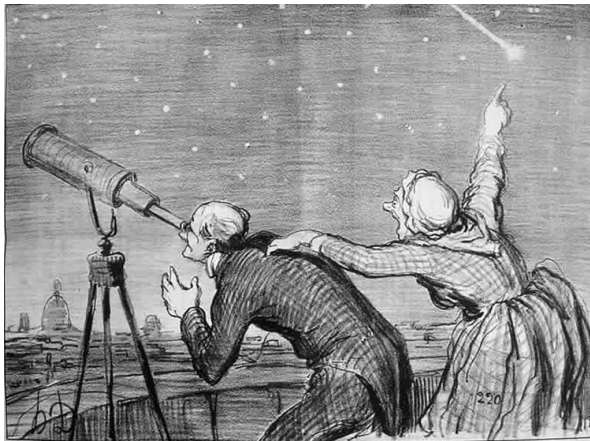
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Applications of ML in the geosciences

References

Unfortunate Monsieur Babinet's inappropriate inductive bias



Honoré Daumier, 1858. cseligman.com/.../daumier.htm

“Monsieur Babinet prévenu par sa portière de la visite de la comète”, translated to ML talk: *You better chose an appropriate hypothesis space!*

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Training and test data

Never evaluate the predictive power of a model on the data it learned from!

- The available data $\mathcal{D}$ is split in two:
 - **Training set**: The examples the algorithm sees and learns from; it optimises $\hat{\theta}$ on these.
 - **Test set**: Held-out examples the model never sees during training; used only once at the end to estimate real-world performance.
- A model that performs well on training data but poorly on test data has **overfit**: it memorised the training examples rather than learning the underlying pattern f .
- A model that performs poorly on *both* has **underfit**: $\mathcal{H}$ was too restricted to capture f .
- The gap between training and test performance is the key diagnostic:

$$\underbrace{J_{\text{test}} - J_{\text{train}}}_{\text{generalisation gap}} \approx 0 \implies \text{Model generalises well.}$$

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

EXAMPLE Permafrost prediction

Common choices of $\mathcal{L}$ depend on the task:

- **Classification:** accuracy, F1-score, e.g. predicting permafrost presence/absence
- **Regression:** Mean Squared Error, e.g. predicting active layer thickness in metres

BEWARE Spatial tasks: Standard metrics ignore spatial and temporal autocorrelation.

If permafrost is rare in the training area, a model predicting “absent” everywhere scores high accuracy but is scientifically useless. *The choice of $\mathcal{L}$ must reflect what matters for the task.*

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Optimisation: How to find the best model (fast)?

Optimisation is the algorithm that searches the hypothesis space $\mathcal{H}$ for the lowest-loss model $\hat{f}$ and returns it.

Given a loss $\mathcal{L}$ and a hypothesis space $\mathcal{H}$, optimisation finds the parameters $\hat{\theta}$ that minimise $\mathcal{L}$. Searching $\mathcal{H}$ is operationalised as:

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(\underbrace{\hat{f}(X; \theta)}_{\hat{y}}, y). \quad (5)$$

The search strategy depends on the structure of $\mathcal{H}$:

- **Analytical computation** when closed-form solution exists (e.g. ordinary least squares for linear models: $\hat{\theta} = (X^\top X)^{-1} X^\top y$).
- Numerical **gradient descent**: When no closed form exists, iteratively descend the slope of $\mathcal{L}$ w.r.t. θ (e.g. neural networks, $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$).
- **Combinatorial search**: Greedy discrete search when no gradients exist (e.g. splitting rules in decision trees)

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Optimisation

Computational tractability of optimization is about whether the computer can find the best $\hat{f}$ within $\mathcal{H}$ efficiently.

Even a highly expressive $\hat{f}$ is only useful in practice if the computer can **efficiently** find the best parameters for it.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

[Domingos, 2012; Goodfellow et al., 2016]. Later we'll see that Bayesians do not optimise at all: They do not just output one “best” solution, but all of them, weighted by how probable they are.

A taxonomy of models

ML algorithms

- 'Traditional ML'
- Deep learning

Key distinction: Traditional ML relies on hand-crafted features fed into a shallow model, while deep learning learns hierarchical representations end-to-end from raw data.

Applicability	Trad. ML	Deep Learning
Tabular, structured data	+	0
Images, waveforms, grids	0	+
Small datasets	+	−
Interpretability matters	+	0

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

A taxonomy of models

Learning from static data (instant supervision signal/"feedback")

- Targets/labels available: Supervised learning
- No targets/labels available: Unsupervised learning

Learning from interaction (delayed "feedback")

- Reinforcement learning (not covered here)

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

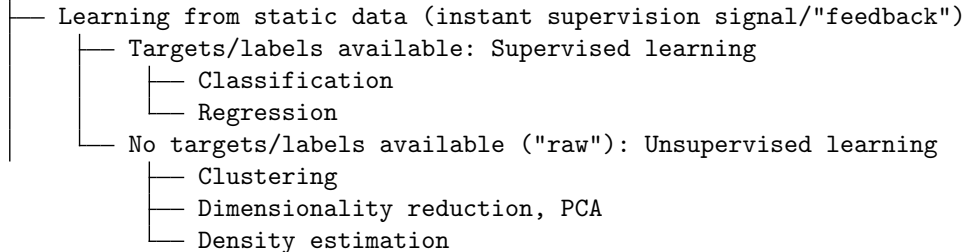
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Applications of ML in the geosciences

References

A taxonomy of models

Machine Learning



Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

**Algorithm
taxonomies**

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

A taxonomy of models: (Non-)parametric models

Numerical models

- Non-parametric models: Number of parameters grow with the data.
- Parametric models: Number of parameters fixed by model architecture.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

[Wasserman, 2004]

EXAMPLE Parametric vs. non-parametric models

Parametric models	Non-parametric models
Assume a specific functional form. Number of parameter model-specific and fixed. Easier to interpret, more prone to underfitting.	Make fewer assumption. Number of parameters (complexity) grows with data. Flexible, more prone to overfitting.
EXAMPLES for prediction, classification, and density estimation:	
Linear regression	Non-parametric curve estimation, smoothing (splines)
Logistic regression, naive Bayes	k -nearest neighbour (k -NN)
Parametric statistical distribution (e.g., Gaussian or Gaussian mixture)	Histogram or kernel density estimation (KDE)

RECALL: Truly assumption-free ML is impossible. Every model embeds assumptions somewhere, even if implicitly.

QUESTION: Are ANNs parametric or not?

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

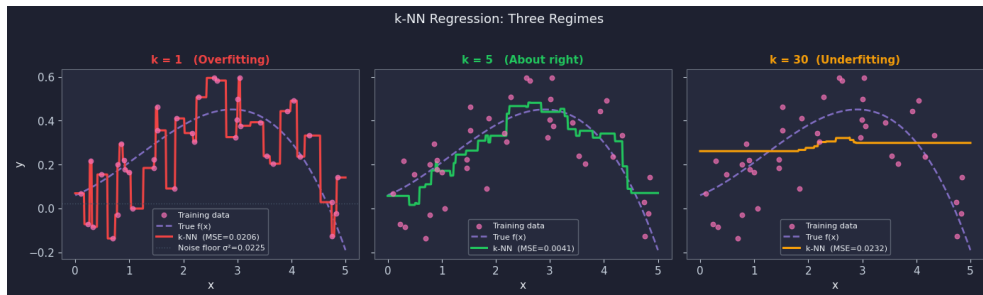
References

Bias–variance trade-off and Underfitting–overfitting

We explore this numerically by means of a concrete example: k -NN regression.

Error decomposition:

$$\text{Total error} = (\text{Bias})^2 + \text{Variance} + \text{Noise} \quad (6)$$



Own work.

Code: `bias-variance-knn.ipynb`

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

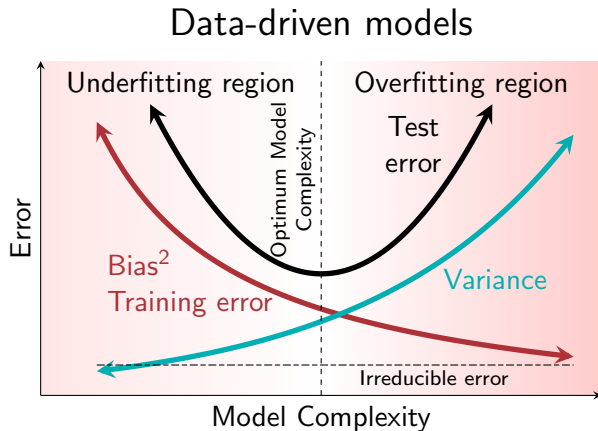
Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Bias–variance trade-off: ML models



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

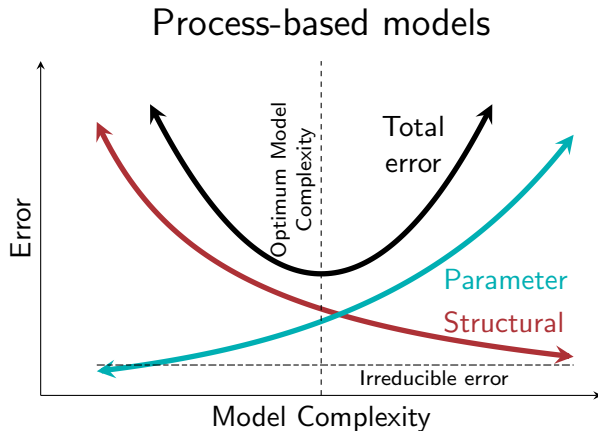
Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Bias–variance trade-off: Physical models



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Bias–variance trade-off? Not always: Double descent

The classical bias–variance trade-off predicts a single optimal model complexity, beyond which overfitting increases test error monotonically. **This does not hold for modern ANNs.**

Belkin et al. [2019] describe two regimes separated by the **interpolation threshold**, where the number of parameters $p \approx$ number of training samples n :

1. **Underparameterised regime** ($p \ll n$): The classical bias–variance trade-off holds, the test error has a single U-shaped minimum.
2. **Overparameterised regime** ($p \gg n$): beyond the threshold, further increasing model size *reduces* test error again, often below the classical optimum.

Near the interpolation threshold ($p \approx n$) generalisation is worst. Large overparameterised networks avoid this by sitting deep in the overparameterised regime, where implicit regularisation drives good generalisation.

Belkin et al. [2019]

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

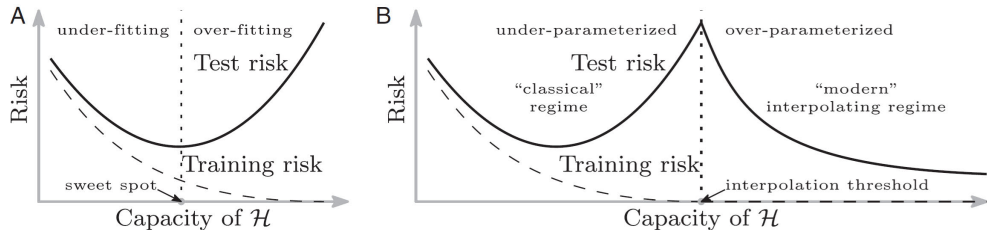
5 schools in ML

Applications of ML in the geosciences

References

Bias-variance trade-off? Not always: Double descent

The interpolation threshold is reached when the number of parameters $\approx$ number of training samples. Large overparameterised networks sit in the modern regime, where implicit regularisation from gradient descent drives good generalisation.



Belkin et al. [2019]

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias-variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Belkin et al. [2019]; see <https://www.youtube.com/watch?v=z64a7USuGX0>.

EXAMPLE Varying complexity of snow models

Briefly compare the parsimonious semi-empirical Δ_{SNOW} [Winkler et al., 2021] model with the multi-physics/thermodynamic Crocus snowpack model [Vionnet et al., 2012] in terms of (i) number of model parameters and (ii) required meteorological input/driving variables.

Links: [> \$\Delta\$ SNOW model](#); [>CROCUS model](#).

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Similar in terms of what?

All ML algorithms rely on some notion of 'distance' to assess and discriminate outcomes for the optimisation. Another, more intuitive, example is the **k-NN** algorithm: Nearest in terms of what?

We discuss distances and why they are tricky in large-dimensional spaces.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Distances and metrics

The Minkowski distance of order k is defined as

$$\ell_k(p, q) := \left(\sum_{i=1}^d c_i |p_i - q_i|^k \right)^{1/k} \quad (7)$$

of the d -dimensional space. c_i is a weight factor that can be (cautiously) used to scale the data in different dimensions. When might c_i be important?

A **metric** is a distance measure $\ell_k(p, q)$ that satisfies the following conditions:

- **Positivity:** $\ell_k(p, q) \geq 0$.
- **Identity:** $\ell_k(p, q) = 0 \iff p = q$
- **Symmetry:** $\ell_k(p, q) = \ell_k(q, p)$
- **Triangle inequality:** $\ell_k(p, r) \leq \ell_k(p, q) + \ell_k(q, r)$

The Minkowski distance satisfies all these conditions for $k \geq 1$.

[Deisenroth et al., 2020, p. 62]

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

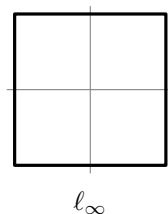
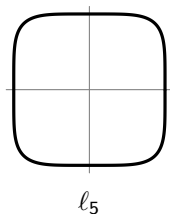
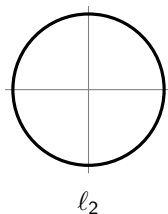
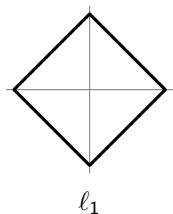
Applications of ML in the geosciences

References

Only ℓ_2 circles are round

Important special cases of the Minkowski distance:

- ℓ_1 distance: Taxicab/Manhattan distance measures distance as the sum of the distances in each coordinate.
- ℓ_2 distance: Euclidian distance
- ℓ_∞ distance: the Chebyshev distance as the limiting case of $p \rightarrow \infty$ measures distance as the maximum of the distances in each coordinate



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias-variance trade-off

Distances and metrics

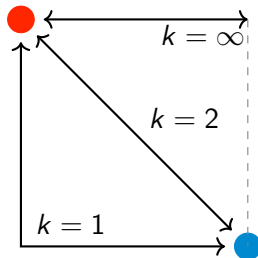
Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Minkowski distances



Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias-variance
trade-off

**Distances and
metrics**

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

EXERCISE Distance-dependent model preference

RECALL Choosing $\mathcal{L}$ encodes your scientific priorities and can affect the model ranking. Here, we illustrate this.

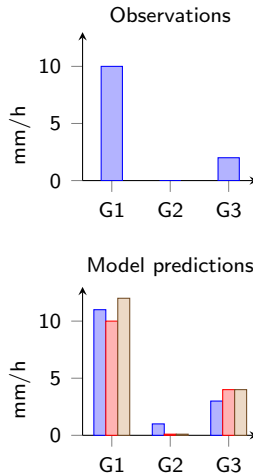
3 rain gauges are placed in a catchment. At a given hour, you observe rainfall (mm/h) $y = (10, 0, 2)$. Three models produce predictions:

$$\hat{y}_{M1} = (11, 1, 3), \quad \hat{y}_{M2} = (10, 0, 4), \quad \hat{y}_{M3} = (12, 0, 4).$$

You also have uncertainty information: gauges 1 and 3 are affected by the same storm cell, so their errors are correlated,

$$\Sigma = \begin{pmatrix} 9 & 0 & 8 \\ 0 & 0.25 & 0 \\ 8 & 0 & 9 \end{pmatrix}.$$

Calculate and interpret the ℓ_1 , ℓ_2 , ℓ_∞ distances, and the Mahalanobis distance.



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

SOLUTION Distance-dependent model preference

Residuals $e_k = \hat{y}_{Mk} - y$ with $y = (10, 0, 2)$:

$$e_1 = (1, 1, 1), \quad e_2 = (0, 0, 2), \quad e_3 = (2, 0, 2)$$

Metric	M_1	M_2	M_3	Winner
ℓ_1	3	2	4	M_2
ℓ_2	$\sqrt{3} \approx \mathbf{1.73}$	2	$2\sqrt{2} \approx 2.83$	M_1
ℓ_∞	1	2	2	M_1

- M_1 spreads a small error across all gauges $\Rightarrow$ no single bad gauge, so ℓ_∞ (worst-case) and ℓ_2 prefer it.
- M_2 has *one localised error* and is exact elsewhere $\Rightarrow$ small *total* error, ℓ_1 prefers it.
- M_3 is dominated: is never preferred under any metric.

Upshot: If total rainfall volume matters, use ℓ_1 (catchment water balance). If every gauge must be reliable, penalise the worst-case error with ℓ_∞ (flood early warning).

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Distance-dependent model preference

Different distance measures encode different scientific priorities (robustness, worst-case risk, correlated uncertainty, or information content), and can legitimately lead to different conclusions about which model is “best”.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Large-dimensional oranges have virtually no pulp.

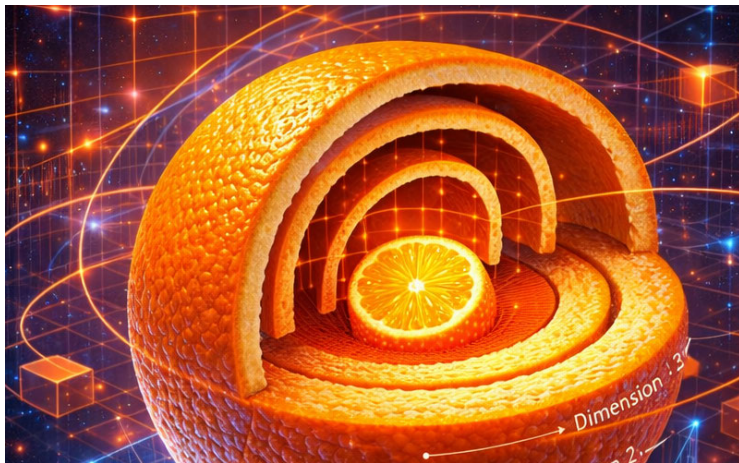


Image generated using AI.

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Non-intuitive properties of high-dimensional spaces

Large-dimensional spaces are vast and tend to be **empty, regular, and concentrated** along the boundaries.

These non-intuitive properties of large-dimensional spaces poses challenges to ML algorithms, collectively known as '**Curse of dimensionality**':

- For instance-based learning: Data requirement grows exponentially with number of features (dimension) d . “The data has seen nothing.”
- For sampling and search algorithms: Data points in high-dimensional spaces data points tend to be concentrated near the surface or boundary of the space. Algorithms might waste computational resources to search in empty spaces.
- For k -NN: Relative distances between points collapse, making k -NN prone to noise and reducing the discriminative power.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

EXERCISE Deploying ground surface temperature (GST) sensors

The ground surface temperature (GST) reflects the heat exchange between ground and atmosphere (including the effects of snow cover) and is an important variable to characterize the ground thermal regime and the absence/presence of permafrost. In complex mountain terrain, GST strongly varies as a function primarily of geographic position ($\rightarrow$ dominant weather patterns), altitude ($\rightarrow$ air temperature), and slope/aspect ($\rightarrow$ insolation); and secondarily of the local terrain shape ($\rightarrow$ duration and thickness of snow cover) and substrate type ($\rightarrow$ affecting local heat transfer). Estimate how many GST sensors are theoretically necessary to cover the Tyrolean Alps. For that, you need to discretize the feature space, for example the elevation in bands of 100 m. Discuss reasonable feature space discretisations. Are all combinations of feature values similarly likely to occur in our real world? Estimate:

1. How many GST sensors are theoretically needed to cover the feature space?
2. How many GST sensors are reasonably needed in a real landscape?

Topics: Combinatorial explosion, manifolds.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

EXERCISE Pairwise distances in the unit hypercube

Be X, Y two independent variables, with uniform distribution on the $[0, 1]^d$ unit hypercube. Numerically compute histograms of pairwise distances between $n = 100$ points sampled uniformly in the hypercube $[0, 1]^d$. Briefly comment on how the mean ℓ_2 distance and its standard deviation behaves as a function of number of dimensions d , and what this means for the notion of “nearest neighbour”.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

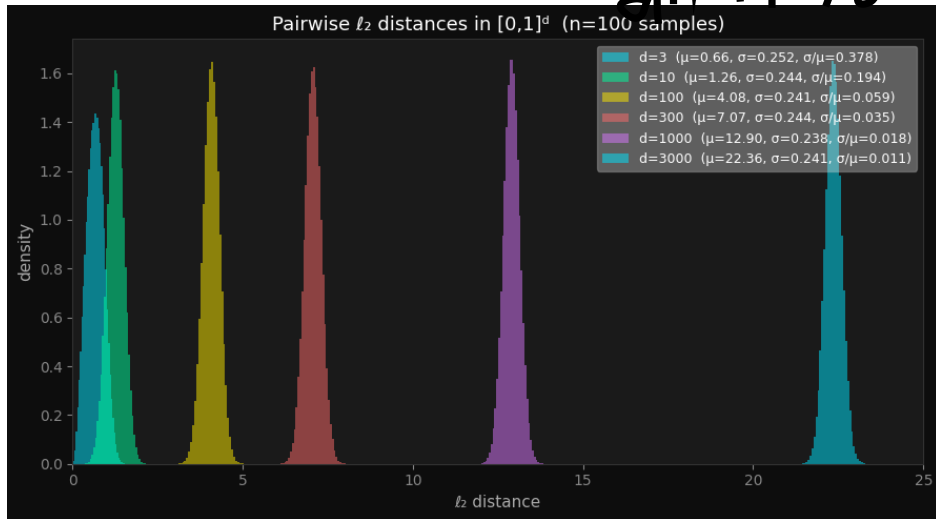
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Applications of ML in the geosciences

References

Pairwise ℓ_2 distances in the $[0, 1]^d$ hypercube

$d \uparrow, n \uparrow$



Own work.

.ipynb

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Distances in high-dimensional spaces (CoD revisited)

Although distances grow with dimension, their relative variability vanishes: almost all points are at approximately the same distance from each other. As a result, the nearest neighbour does not significantly differ from any other neighbour, $d_{max}/d_{min} \xrightarrow{d \rightarrow \infty} 1$, making ℓ_2 -based nearest-neighbour methods increasingly unstable and sensitive to noise in high dimensions.

Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Classical ways to avoid the CoD

- Parsimonious models: Reduce the number of model parameters.
- Dimension reduction: Reduce the number of features.
- Regularisation: Add additional constraints to regularize the model.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Non-intuitive properties of high-dimensional spaces

Large-dimensional spaces are vast and tend to be **empty, regular, and concentrated** along the boundaries.

These non-intuitive properties of large-dimensional spaces can be harnessed by ML algorithms, collectively known as '**Blessing of dimensionality**':

- Hard-to-separate data can be “elevated” to higher-dimensional spaces where linear separation is trivial: **SVM**.
- Local optima among many almost-equally good solutions might be good enough; no need to search exhaustively for the one best global optimum: **Good solutions are abundant in overparameterized models** (→ double descent).
- Real data lives in specific areas of the feature spaces (called manifold), because it happens that our universe seems to be governed by some rules.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

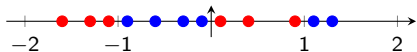
5 schools in ML

Applications of ML in the geosciences

References

Harnessing high-dim spaces: Kernel mapping

Original space x



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

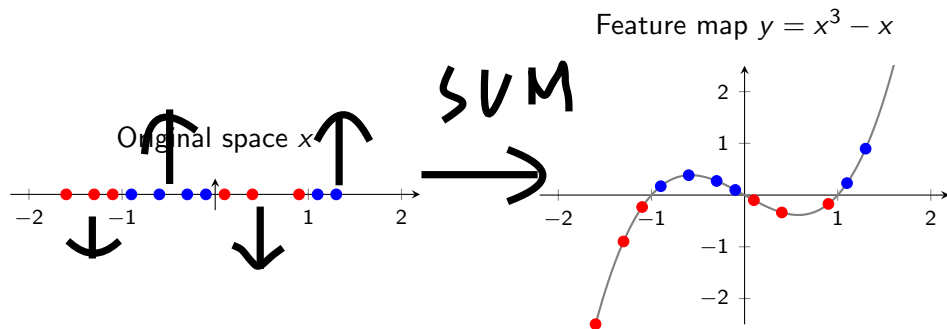
Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Harnessing high-dim spaces: Kernel mapping



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

The Five Schools of Machine Learning

- ML is divided into five distinct *epistemic* schools of thought, each rooted in a different scientific tradition.
- Each school defines its own:
 - **Representation** — how knowledge is stored
 - **Learning** — how generalisation happens
 - **Assumptions** — what is baked in
- Neural networks currently dominate, but are *not* the only form of AI.

The Holy Grail

Unifying insights and algorithms across all five schools is widely regarded as the “master algorithm” — a general learner and arguably a necessary step towards AGI.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

The Five Schools of Machine Learning

School	Origin	Core Idea	Master Algorithm
Symbolists	Logic, philosophy	Rules, logic, decision trees	Inverse deduction
Bayesians	Statistics	Probabilistic and graphical models	Belief propagation
Connectionists	Neuroscience	Artificial neural networks (ANNs)	Backpropagation
Evolutionaries	Evolutionary biology	Genetic algorithms, selection	Genetic programming
Analogizers	Psychology, cognitive science	Similarity- and kernel-based methods	k -NN, SVM

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

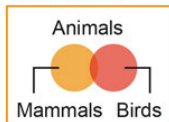
Applications of ML in the geosciences

References

Based on P. Domingos, *The Master Algorithm* (2015).

The five schools in ML

Symbolists



Use symbols, rules, and logic to represent knowledge and draw logical inference

Favored algorithm
Rules and decision trees

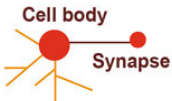
Bayesians



Assess the likelihood of occurrence for probabilistic inference

Favored algorithm
Naive Bayes or Markov

Connectionists



Recognize and generalize patterns dynamically with matrices of probabilistic, weighted neurons

Favored algorithm
Neural networks

Evolutionaries



Generate variations and then assess the fitness of each for a given purpose

Favored algorithm
Genetic programs

Analogizers



Optimize a function in light of constraints ("going as high as you can while staying on the road")

Favored algorithm
Support vectors

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Source: Pedro Domingos, *The Master Algorithm*, 2015

Figure slightly modified from rajivsworldlife, originally from a (now irretrievable) PWC article. Idea by Domingos [2018].

[Domingos, 2018]

Comparison of the ML schools*

Core idea, typical algorithms, strengths, weaknesses, application domains (evolutionaries excluded):

Symbolists	Connectionists	Bayesians	Analogizers
Rules and logical structures	Weighted networks of artificial neurons	Probabilistic inference	Similarity-based reasoning
Decision trees C4.5 , rule sets, logic programs	Neural networks, deep learning CNNs RNNs	Bayesian networks, DAGs, Markov networks, MCMC Naive Bayes	k-NN , SVMs , kernel methods (instance-based, “lazy” learning)
Interpretability, structured knowledge, use of prior knowledge	Flexible, high predictive performance, scaleable	Principled uncertainty handling, priors compensate sparse data	Strong theory, flexible decision boundaries
Brittle, sensitive to noise, struggles with contradicting data	Opaque, data- and compute-hungry	Computationally demanding, sensitive to prior	Sensitive to feature scaling, costly at inference, memory-hungry
Domains with explicit rules and structure	(Satellite) images, large-scale data	Scientific modelling, uncertainty-aware decisions	Small-to-medium data, strong similarity structure

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Symbolists

Core idea: Learning = inverse deduction

- Knowledge represented as **rules and logic** (symbolic structures)
- Learn by working *backwards* from data to the rules that could have produced it
- Representations: decision trees, if-then rules, first-order logic
- Strength: interpretable, no black box, strong use of prior knowledge
- Weakness: brittle with noise; hard to scale

Origin: Logic and philosophy

Example

A decision tree for spam detection:

IF "buy now" $\in$ subject
AND sender unknown
THEN spam

Each rule was *induced* from labelled examples via inverse deduction.

Master Algorithm

Inverse deduction (inductive logic programming)

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias-variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

ELIZA chatbot (1966)

Welcome to

```

EEEEEE LL IIII ZZZZZZ AAAAA
EE LL II ZZ AA AA
EEEE LL II ZZZ AAAAAA
EE LL II ZZ AA AA
FFFFFF LLLLLL IIII ZZZZZZ AA AA

```

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

```
ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

Wikipedia/public domain.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Bayesians

Core idea: Learning = probabilistic inference under uncertainty

- All knowledge expressed as **probability distributions**
- Update beliefs when new evidence arrives:
 $P(H|D) \propto P(D|H) P(H)$
- Representations: Bayesian networks, graphical models, HMMs
- Strength: principled handling of uncertainty; works with little data
- Weakness: choosing priors; computationally expensive

Origin: Statistics

Example

Spam filter updated live:

Prior: 20% of emails are spam.

Evidence: email contains "lottery".

Posterior: probability jumps to 95%.

Every new email *updates* the model.

Master Algorithm

Belief propagation (sum-product algorithm)

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

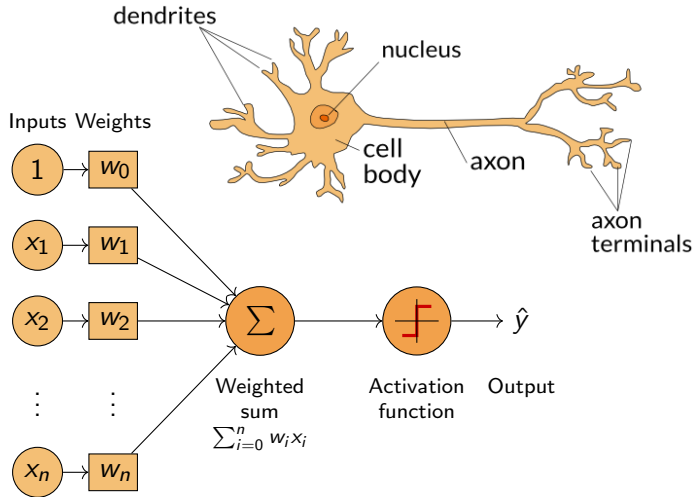
Large-dim. spaces

5 schools in ML

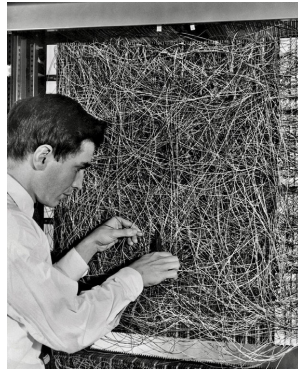
Applications of ML in the geosciences

References

The perceptron: Building block of neural networks (1957)



Frank Rosenblatt with a Mark I Perceptron computer in 1960.



Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

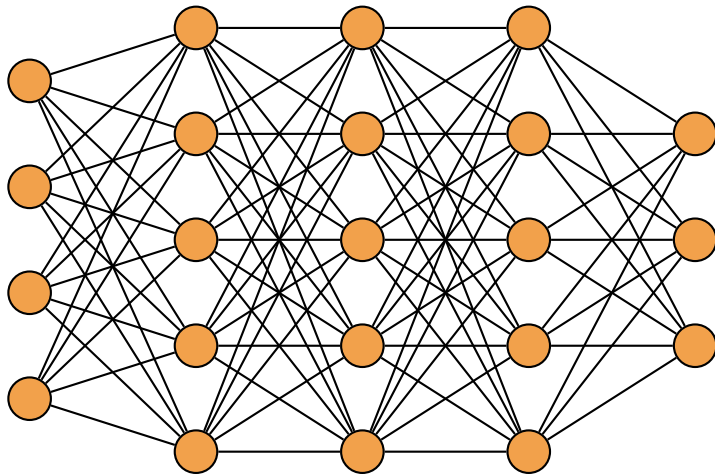
Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Artificial neural networks



Motivating
example

Definition and
aims of ML

Representation and
inductive bias

Evaluation

Optimisation

Algorithm
taxonomies

Bias–variance
trade-off

Distances and
metrics

Large-dim. spaces

5 schools in ML

Applications of ML
in the geosciences

References

Analogizers

Core idea: Learning = reasoning by similarity, comparing new problems to stored examples

- Knowledge stored *implicitly* in the training examples themselves
- Classify new input by finding the most similar stored examples
- Representations: k -NN, SVMs, kernel methods, case-based reasoning
- Strength: flexible; no explicit model needed
- Weakness: memory-hungry, slow at inference; sensitive to feature scaling, “curse of dimensionality”

Origin: Cognitive psychology

Example

Recommender system:

“Users who bought the same 5 books as you also bought X .”

SVMs find the *maximum-margin hyperplane* separating classes using only a handful of *support vectors* (the most similar boundary examples).

Master Algorithm

k -NN, **Support vector machines** (kernel trick)

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

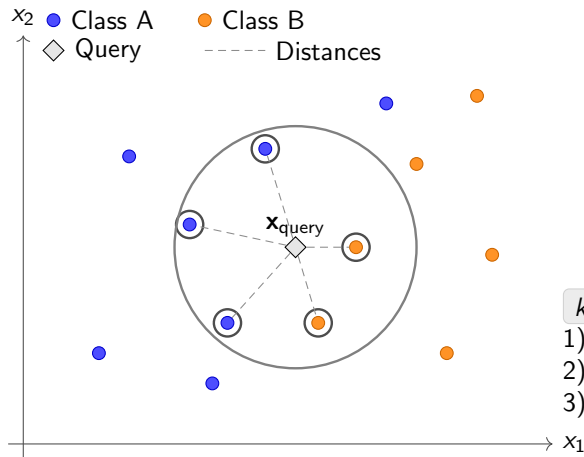
Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

k -nearest neighbour algorithm k -NN



k -NN ($k = 5$)

- 1) Compute distances
- 2) Pick k closest points
- 3) Majority vote $\rightarrow$ Label

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias-variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

ML in the geosciences

Dramsch [2020] classified ML applications by *capability gain*:

- I. Build a surrogate ML model of a well-understood process, usually to cut computational costs (gains in speed).
- II. Build an ML model of a task previously only possible with human interaction, usually to automate tasks (much more gains in speed).

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

[Dramsch, 2020; Sun et al., 2022; Zhao et al., 2024]. [EXAMPLES for each class \(next slides\)](#).

Class I: ML model emulation

- ML surrogate models

EXAMPLES Computing glacier flow of the Alps during the last Glacial (calculation takes minutes on consumer-grade computer instead of days on a supercomputer).

Jouvet & Cordonnier [2023]; Leger et al. [2025]

<https://www.youtube.com/watch?v=IbL0Fh3U9gI>

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance
trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

Class II: Automation of a human interpretive step

- Extracting information from unstructured data sets, notably aerial or satellite imagery

EXAMPLES Mapping rooftop water tanks, outlining landforms like glaciers or rock glaciers. Retrieving albedo information. Clustering geophysical data [Mathys et al., 2025].

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

READING TASK AND RECAP

We'll have a case study on statistical permafrost distribution modelling. As a preparation, read (or skim) Pandey et al. [2023] and briefly answer the following questions:

1. Classify their features based on the data types introduced in the lecture (observational, model-derived, synthetic, direct, proxy).
2. Which type of data is used as prediction and target variables, and what is their relation?
3. Do you see potential methodological issues due to the used data? How were the results verified?
4. After pruning the features using RFE, how do the “surviving” features compare with permafrost decision tree introduced in the lecture?
5. Which of the 10 used classifiers are you already familiar with?

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

HOMEWORK TASK & PRESENTATION (2')

What are typical applications of ML in the geosciences?

In groups of 2, find 2 geoscience papers that use ML and address the questions below. The search space includes the entire field of AI (not only traditional ML in the restricted sense), but find papers of two different schools. Briefly present:

- Which affiliation and scientific domain the authors belong to;
- Which geoscientific question or problem was addressed in the paper;
- Which algorithms were used to solve the tasks, and which ML school they belong to;
- Whether (or to what extent) the problem had been solvable without ML/AI, large data sets, and modern computing power.

Brief oral presentation without slides.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

References I

- Belkin, M., Hsu, D., Ma, S., & Mandal, S. (2019). Reconciling modern machine-learning practice and the classical bias-variance trade-off. *Proceedings of the National Academy of Sciences*, 116(32), 15849–15854.
- Borges, J. L. (1944). Funes, the Memorious [Funes el memorioso]. In *Ficciones*. Argentina: Editorial Sur.
- Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). *Mathematics for Machine Learning* (1 ed.). Cambridge University Press.
- Domingos, P. (2012). A few useful things to know about machine learning. *Communications of the ACM*, 55(10), 78–87.
- Domingos, P. (2018). *The Master Algorithm: How the Quest for the Ultimate Learning Machine Will Remake Our World*. New York: Basic books (Perseus Book Group).
- Dramsch, J. S. (2020). 70 years of machine learning in geoscience in review. In *Machine Learning in Geosciences*, volume 61 of *Advances in Geophysics* (pp. 1–55). Elsevier.
- Ebohon, B. & Schrott, L. (2008). Modeling mountain permafrost distribution. a new permafrost map of Austria. In *Proceedings of the Ninth International Conference on Permafrost, Fairbanks, Alaska*, (pp. 397–402).
- Etzelmüller, B., Hoelzle, M., Flo Heggem, E. S., Isaksen, K., Mittaz, C., Mühl, D. V., Ødegård, R. S., Haeberli, W., & Sollid, J. L. (2001). Mapping and modelling the occurrence and distribution of mountain permafrost. *Norsk Geografisk Tidsskrift - Norwegian Journal of Geography*, 55(4), 186–194.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
- Haeberli, W. (1975). *Untersuchungen zur Verbreitung von Permafrost zwischen Flüelapass und Piz Grialetsch (Graubünden)*, volume 17 of *Mitteilungen der Versuchsanstalt für Wasserbau, Hydrologie und Glaziologie, ETH Zürich*. Zürich.
- Hoelzle, M., Mittaz, C., Etzelmüller, B., & Haeberli, W. (2001). Surface energy fluxes and distribution models of permafrost in European mountain areas: an overview of current developments. *Permafrost and Periglacial Processes*, 12(1), 53–68.
- James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning: with Applications in Python*. Springer Texts in Statistics. Cham: Springer International Publishing.
- Jouvet, G. & Cordonnier, G. (2023). Ice-flow model emulator based on physics-informed deep learning. *Journal of Glaciology*, 69(278), 1941–1955.
- Keller, F. (1992). Automated mapping of mountain permafrost using the program PERMAKART within the geographical information system ARC/INFO. *Permafrost and Periglacial Processes*, 3(2), 133–138.
- Leger, T. P. M., Jouvet, G., Kamleitner, S., Mey, J., Herman, F., Finley, B. D., Ivy-Ochs, S., Vieli, A., Henz, A., & Nussbaumer, S. U. (2025). A data-consistent model of the last glaciation in the Alps achieved with physics-driven AI. *Nature Communications*, 16(1), 848.

Motivating example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References

References II

- Mathys, T., Azimshoev, M., Bekturysunov, Z., Hauck, C., Hilbich, C., Duishonakunov, M., Kayumov, A., Kassatkin, N., Kapitsa, V., Martin, L. C. P., Mollaret, C., Navruzshoev, H., Pohl, E., Saks, T., Silmonov, I., Musaev, T., Usubaliev, R., & Hoelzle, M. (2025). Quantifying permafrost ground ice contents in the Tien Shan and Pamir (Central Asia): a petrophysical joint inversion approach using a geometric mean model. *The Cryosphere*, 19(12), 6591–6628.
- Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill Series in computer Science. New York: McGraw-Hill.
- Otto, J.-C., Schrott, L., & Keller, F. (2020). Map of permafrost distribution for Austria, Europe [data set].
- Pandey, A., Yadav, B. C., Wani, J. M., & Dimri, A. P. (2023). Permafrost estimation model in Upper Indus Basin. *Journal of Earth System Science*, 132(4), 156.
- Schrott, L., Otto, J.-C., & Keller, F. (2012). Modelling alpine permafrost distribution in the Hohe Tauern region, Austria. *Austrian Journal of Earth Sciences*, 105(2), 169–183.
- Sun, Z., Sandoval, L., Crystal-Ornelas, R., Mousavi, S. M., Wang, J., Lin, C., Cristea, N., Tong, D., Carande, W. H., Ma, X., Rao, Y., Bednar, J. A., Tan, A., Wang, J., Purushotham, S., Gill, T. E., Chastang, J., Howard, D., Holt, B., Gangodagamage, C., Zhao, P., Rivas, P., Chester, Z., Orduz, J., & John, A. (2022). A review of Earth Artificial Intelligence. *Computers & Geosciences*, 159, 105034.
- Vionnet, V., Brun, E., Morin, S., Boone, A., Faroux, S., Le Moigne, P., Martin, E., & Willemet, J.-M. (2012). The detailed snowpack scheme Crocus and its implementation in SURFEX v7.2. *Geoscientific Model Development*, 5(3), 773–791.
- Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. Springer Texts in Statistics. New York, NY: Springer New York.
- Winkler, M., Schellander, H., & Gruber, S. (2021). Snow water equivalents exclusively from snow depths and their temporal changes: the Δ snow model. *Hydrology and Earth System Sciences*, 25(3), 1165–1187.
- Wolpert, D. H. (1996). The Lack of A Priori Distinctions Between Learning Algorithms. *Neural Computation*, 8(7), 1341–1390.
- Zhao, T., Wang, S., Ouyang, C., Chen, M., Liu, C., Zhang, J., Yu, L., Wang, F., Xie, Y., Li, J., Wang, F., Grunwald, S., Wong, B. M., Zhang, F., Qian, Z., Xu, Y., Yu, C., Han, W., Sun, T., Shao, Z., Qian, T., Chen, Z., Zeng, J., Zhang, H., Letu, H., Zhang, B., Wang, L., Luo, L., Shi, C., Su, H., Zhang, H., Yin, S., Huang, N., Zhao, W., Li, N., Zheng, C., Zhou, Y., Huang, C., Feng, D., Xu, Q., Wu, Y., Hong, D., Wang, Z., Lin, Y., Zhang, T., Kumar, P., Plaza, A., Chanussot, J., Zhang, J., Shi, J., & Wang, L. (2024). Artificial intelligence for geoscience: Progress, challenges, and perspectives. *The Innovation*, 5(5), 100691.

Motivating
example

Definition and aims of ML

Representation and inductive bias

Evaluation

Optimisation

Algorithm taxonomies

Bias–variance trade-off

Distances and metrics

Large-dim. spaces

5 schools in ML

Applications of ML in the geosciences

References